Workshop 2: Q-Learning and SARSA on CliffWalking

ELEN90098 Reinforcement Learning for Engineering
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Abstract

We implemented tabular Q-learning, SARSA(0), five-step SARSA, and SARSA(λ) and evaluated them in Gymnasium's CliffWalking-v1. Task 1 established the environment, completed the provided code, and compared one 1,000-episode training run per algorithm in TensorBoard. Task 2 repeated the comparison with five random seeds and varied Q-learning's learning rate α and SARSA(λ)'s trace parameter λ. Q-learning consistently finished with a return of −13, the shortest safe route. SARSA(0) and SARSA(λ=0.1) finished near −17, while five-step SARSA varied more across seeds. With our operational speed measure, SARSA(λ=0.1) reached a return of −25 earliest. All tested α values eventually achieved −13; moderate λ values learned faster than λ=0 or λ=0.9 under the fixed training schedule.

1. Environment and algorithms

CliffWalking is a 4 × 12 grid. The agent starts at (3, 0), aims for (3, 11), and the ten cells (3, 1) through (3, 10) are a cliff. An ordinary move, including the move into the goal, gives reward −1. Entering the cliff gives −100 and returns the agent to the start without ending the episode; entering the goal ends the episode. Actions are discrete: 0 = up, 1 = right, 2 = down, and 3 = left. The environment is finite and discrete. There are 37 nonterminal decision positions (the upper 36 cells and the start). Gymnasium declares Discrete(48) because it encodes all 4 × 12 grid positions; the goal may also appear as a terminal observation, while cliff positions are not occupied after a step.

The agent uses an ε-greedy behavior policy. For a transition (s, a, r, s'), the tabular update is $Q(s,a) \leftarrow Q(s,a) + \alpha\delta$. The algorithms differ in the target used to calculate the temporal-difference error δ:

Algorithm	Target for a nonterminal transition	Key property
Q-learning	$r + \gamma \max_{a'} Q(s',a')$	Off-policy; evaluates a greedy continuation even when the behavior explores.
SARSA(0)	$r + \gamma Q(s',a')$	On-policy; uses the next action actually selected by the ε-greedy policy.
SARSA(n), n=5	$\sum_{i=0}^{n-1} \gamma^i r_{t+i+1} + \gamma^n Q(s_{t+n},a_{t+n})$	Propagates information through five rewards before bootstrapping; the terminal remainder has no bootstrap term.
SARSA(λ)	One-step SARSA error applied to eligible earlier state–action pairs	Uses accumulating traces: $e(s,a) \leftarrow e(s,a)+1$ for the visited pair, then $Q \leftarrow Q+\alpha\delta e$ and $e \leftarrow \gamma\lambda e$.

For a terminal transition, the bootstrap term is zero. The learning discount was $\gamma=0.99$. The reported episode return is the undiscounted sum of rewards, which is distinct from the discounted target used in learning. A direct safe route requires 13 moves, so its undiscounted return is −13.

2. Task 1: CliffWalking

Question 1: State, action, and reward spaces

The finite state and action spaces and the reward and termination rules are described in Section 1. The observation is an integer row × 12 + column; the starting observation is 36 and the goal is 47. Rewards in this task are discrete, with −1 for an ordinary move and −100 for a cliff entry.

Question 2: Initial comparison

We completed the TODO sections in the supplied agent and training script, trained Q-learning, SARSA(0), SARSA(n=5), and SARSA(λ=0.1) for 1,000 episodes each, and compared the TensorBoard episode-return curves.

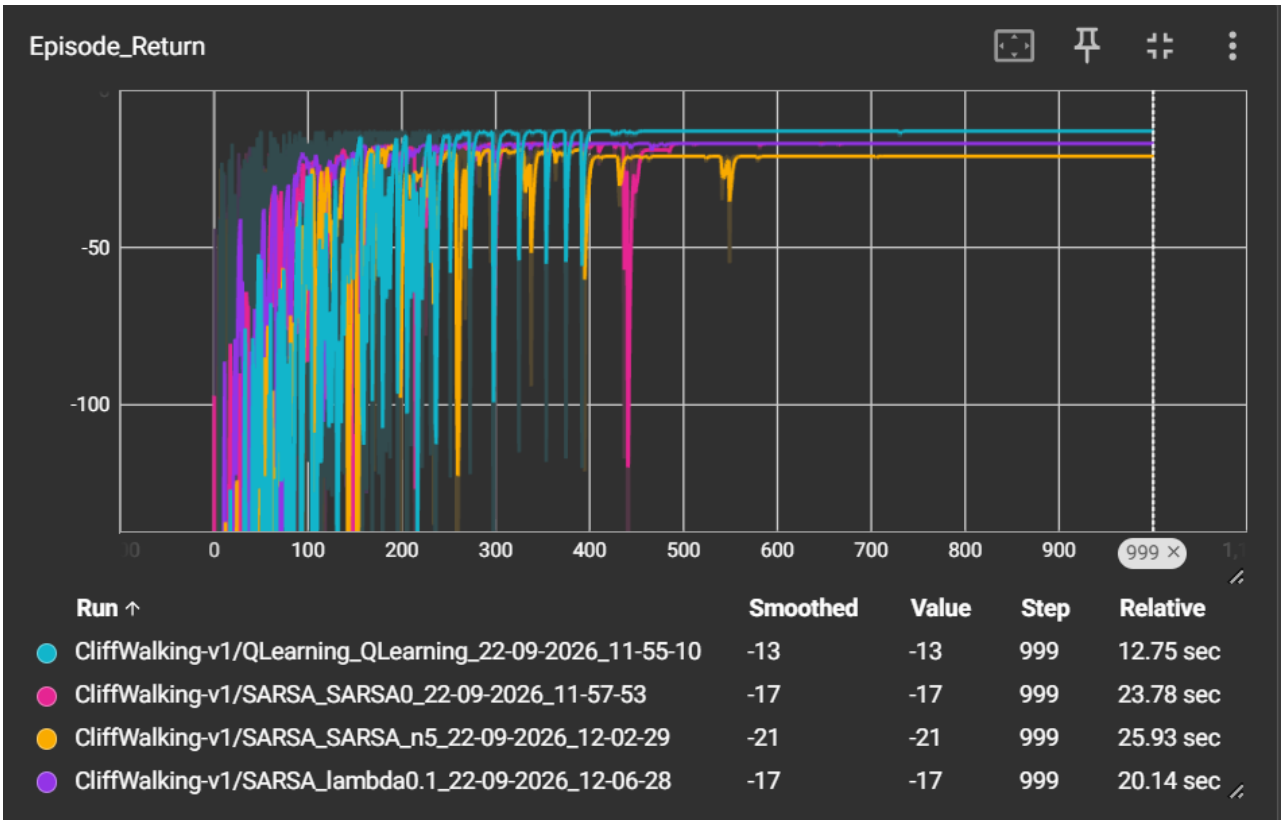


Figure 1. The four Task 1 runs. Values at the final logged step are −13, −17, −21, and −17 for Q-learning, SARSA(0), SARSA(n=5), and SARSA(λ=0.1), respectively. This is a single-run illustration rather than a multi-seed estimate.

All four runs improved sharply from large negative returns caused by exploration and cliff falls. Q-learning reached the shortest safe-route return of −13. The three on-policy runs ended with longer routes in this initial comparison. The Task 2 repetitions below quantify how much of this comparison persists across random seeds; in particular, five-step SARSA's final result varies between runs.

3. Task 2: Experimental design

We used the completed Task 1 QLAgent on CliffWalking-v1. Each distinct configuration was trained for **1,000 episodes with each of five seeds, 0–4**. The common settings were $\gamma=0.99$, initial $\varepsilon=0.5$, episode-wise decay factor 0.99, minimum $\varepsilon=0$, initial Q value 0, and baseline $\alpha=0.5$. The comparisons were:

1. Q-learning, SARSA(0), SARSA(n=5), and SARSA(λ=0.1).
2. Q-learning with $\alpha \in \{0.1, 0.3, 0.5, 0.9\}$.

3. SARSA(λ) with $\lambda \in \{0, 0.1, 0.5, 0.9\}$; $\lambda=0$ is SARSA(0).

Shared settings were run once and reused in the relevant comparisons, giving **nine distinct configurations, 45 runs, and 45,000 training episodes**. A 2,000-step safety cap bounded unusually long episodes; capped episodes were retained in the training results. For each run, we recorded every episode's undiscounted return, the mean of the last 100 training returns, and one final greedy evaluation without exploration or learning. Every final greedy evaluation reached the goal.

To compare learning speed, we report the **first episode at which a 50-episode rolling mean was at least −25 for 20 consecutive rolling windows**. The table gives the median of this episode number across five seeds. This fixed-threshold statistic measures attainment of a useful return; it is not a proof of asymptotic convergence. Figures show the mean of the five smoothed runs with a shaded band of ±1 sample standard deviation.

4. Task 2, Question 1: Four-algorithm comparison

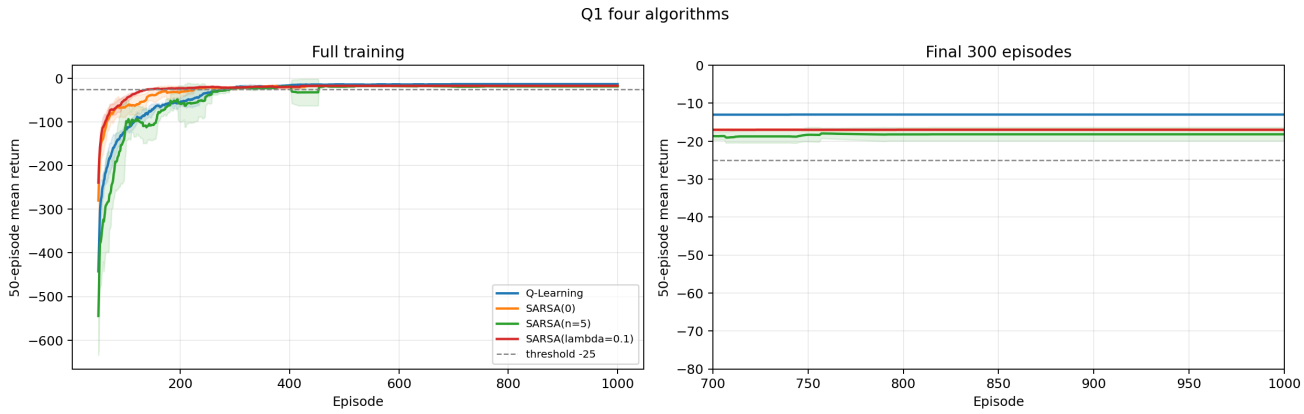


Figure 2. Mean 50-episode rolling returns over five seeds (shading: ±1 standard deviation). The right panel enlarges the final 300 episodes.

Algorithm	Mean return, first 100 episodes	Last-100 return, mean ± SD	Final greedy return, mean ± SD	Median episode reaching −25
Q-learning	−280.0	−13.00 ± 0.00	−13.00 ± 0.00	275
SARSA(0)	−172.2	−17.01 ± 0.02	−17.00 ± 0.00	224
SARSA(n=5)	−326.3	−18.20 ± 1.79	−18.20 ± 1.79	277
SARSA(λ=0.1)	−143.5	−17.00 ± 0.01	−17.00 ± 0.00	**141**

SARSA(λ=0.1) attained the −25 threshold first, followed by SARSA(0). Q-learning ultimately performed best: every seed obtained −13 in both the last 100 training episodes and greedy evaluation. SARSA(0) and SARSA(λ=0.1) finished around −17. Five-step SARSA finished at −17, −21, −17, −19, and approximately −17 across the five seeds, accounting for its larger final variability and for the −21 endpoint in the single Task 1 run.

This behavior is consistent with the update targets. Q-learning bootstraps from the greedy next action and can learn the short route beside the cliff despite exploratory behavior. SARSA methods account for the next action the behavior policy may actually take, so cliff risk during exploration can favor a longer route. Eligibility traces can spread useful feedback to earlier decisions, plausibly explaining the earlier threshold crossing for λ=0.1. Five-step SARSA must wait for several rewards before an update and showed more variation in these runs. These are interpretations of the observed runs, not claims that one method is

always faster. Because ε decays rapidly toward zero, a safe on-policy route learned early can persist through the finite training period.

Five of 5,000 five-step SARSA training episodes reached the 2,000-step cap. They affect early-return averages but none of the five final greedy policies failed to reach the goal.

5. Task 2, Question 2: Q-learning learning rate α

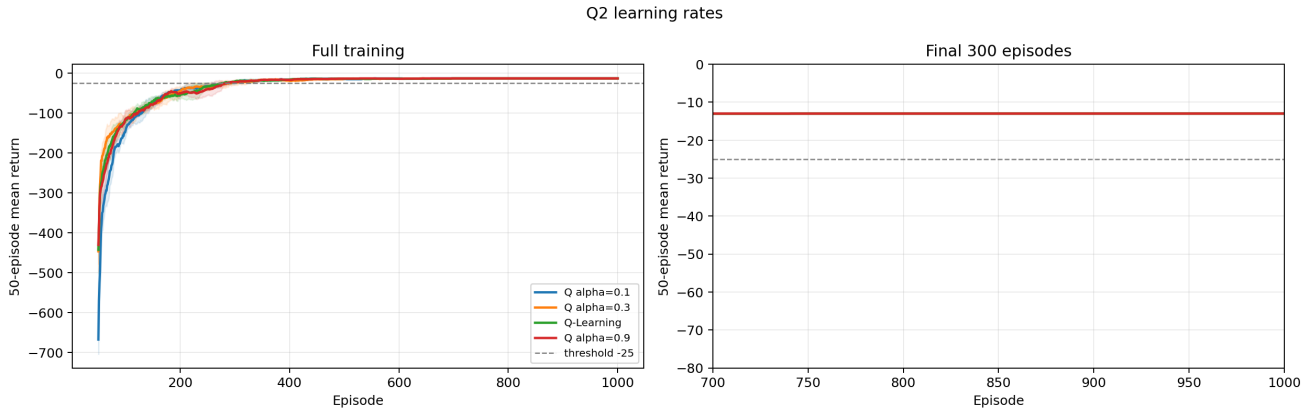


Figure 3. Q-learning with four learning rates; all other settings and seeds are held fixed.

α	Mean return, first 100 episodes	Last-100 return, mean ± SD	Final greedy return	Median episode reaching −25
0.1	−404.9	−13.00 ± 0.00	−13.00	287
0.3	−281.3	−13.00 ± 0.00	−13.00	275
0.5	−280.0	−13.00 ± 0.00	−13.00	275
0.9	−272.4	−13.00 ± 0.00	−13.00	282

A smaller α incorporates each new TD error more slowly, and α=0.1 had the weakest average return in the first 100 episodes. Raising α can transmit new information faster, but the threshold times here are **not monotonic**: α=0.9 had a slightly better early mean than α=0.5, yet its median threshold episode was 282 rather than 275. A large α also gives individual cliff penalties more influence and may increase sensitivity to recent experience; these data do not establish a persistent high-α instability. After 1,000 episodes, all four tested rates yielded the same −13 final policy in all five seeds. Thus α mainly affected the learning transient within this tested range and schedule, rather than the final outcome.

6. Task 2, Question 3: SARSA(λ) trace parameter

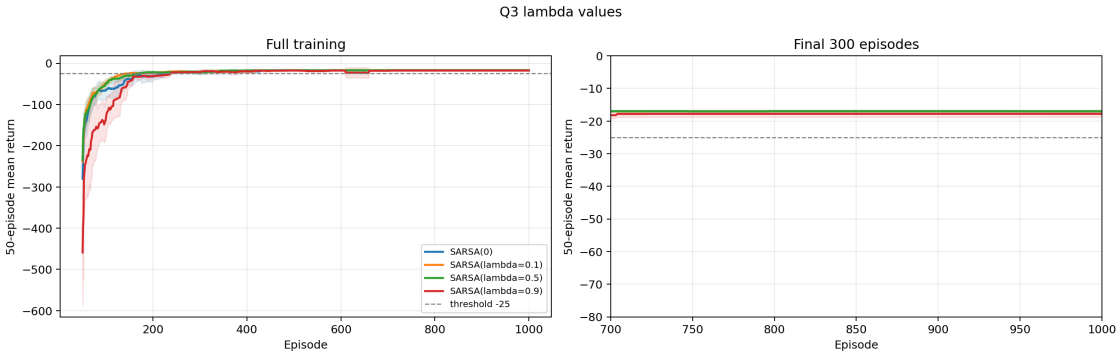


Figure 4. SARSA(λ) with four trace values; λ=0 coincides with SARSA(0).

λ	Mean return, first 100 episodes	Last-100 return, mean ± SD	Final greedy return, mean ± SD	Median episode reaching −25
0	−172.2	−17.01 ± 0.02	−17.00 ± 0.00	224
0.1	−143.5	−17.00 ± 0.01	−17.00 ± 0.00	**141**
0.5	−142.6	−17.00 ± 0.00	−17.00 ± 0.00	171
0.9	−294.8	−17.80 ± 1.09	−17.80 ± 1.10	219

Moderate traces improved learning speed relative to λ=0. Among these values, λ=0.1 reached the threshold earliest, while λ=0.5 also performed well; both finished at −17 in every greedy evaluation. λ=0.9 was slower early and finished at −19 in two seeds and −17 in three. Two of its 5,000 training episodes reached the safety cap. With accumulating traces, a larger λ keeps credit assigned to more previous state–action pairs. This can accelerate propagation when rewards are informative, but it can also spread large cliff penalties or exploratory TD errors over a longer history. The observed λ=0.9 variability is consistent with that trade-off. The best λ depends on the environment, α, ε schedule, and training budget; the present experiment identifies 0.1 as the fastest of the tested values under these conditions.

7. Conclusion and limits

The experiments separate speed of obtaining a useful return from final path quality. SARSA(λ=0.1) first met the −25 threshold, whereas Q-learning alone consistently reached the shortest safe-route return of −13. Changing Q-learning's α altered early learning but not its final result over the tested values. Moderate λ improved SARSA's learning speed, while λ=0.9 gave less reliable final routes in this sample.

The comparison uses five seeds, one deterministic gridworld, a finite 1,000-episode budget, and a rapidly decaying ε. The ±SD bands describe observed variation across runs, not confidence intervals or statistical significance. The threshold is an operational measure, and rare capped episodes make some early averages more negative. A different exploration schedule, training budget, or environment could change the ordering.

Reproducibility materials

- Executed notebook
- Task 2 results archive: per-episode returns, per-run metrics, summaries, and figures
- Task 1 answer and original figure
- Workshop instructions